

ORIGINAL ARTICLE

Telfairia occidentalis-supplemented diet induces changes in sperm parameters and testosterone level in rats

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Summary

The growing patronage on herbal remedies and formulations from natural products in most developing countries has warranted research into certain health challenges including their antifertility effects. This study assessed the effects of boiled *Telfairia occidentalis* (TO) seed-supplemented diets on the level of testosterone and semen quality in Wistar rats. Boiled TO seed diets at 10%, 15% and 30% were given to rats for 60 days. Our study showed that sperm quality was impaired as evidenced by the decreased number of motile spermatozoa, epididymal sperm numbers, percentage live/dead ratio and increased numbers of abnormal spermatozoa comparable to control values ($p < .05$). Feeding of rats with 10% and 15% TO seed-supplemented diets increased testosterone levels nonsignificantly, while in the 30% TO seed diet animals, the level of serum testosterone was found to decrease significantly compared to control values. Furthermore, TO diet caused a nonsignificant increase in the activity of superoxide dismutase and the concentrations of reduced glutathione and malondialdehyde except for the significant increase in malondialdehyde level in the testes of the 10% TO diet group. A nonsignificant decrease in myeloperoxidase activity was also observed in the 10% and 15% but not 30% TO diet group. Histological damages characterised by severe loss of germ cells were more pronounced in the 10% TO diet group. Fourier transform infrared spectroscopy analysis of boiled TO seeds revealed the presence of esters, alkenes, hydroxyl and alcohol functional groups. Thus, boiled TO seed-supplemented diet evoked antifertility effects in rats, and the effects on the toxicity end points investigated were not dose-dependent.

KEYWORDSdiet, sperm quality, *Telfairia occidentalis*, testes, testosterone

1 | INTRODUCTION

The current shift towards the consumption of formulations derived from natural products is necessitated by insufficient therapies to manage certain ailments and social problems (Eboetse, Ikechukwu, Olugbenga, Ayodele, & Caramel, 2011). More so, the desire for regimen with minimal side effects is responsible for the heightened public curiosity in natural products. Thus, recent research paradigm focuses on herbal remedies and dietary food supplements as a means

of solving health challenges and social problems (Eboetse et al., 2011; Eisenberg et al., 1998). However, there is growing experimental evidence indicating that some of these dietary food supplements, fruits, vegetables and seeds have antifertility effects (Abarikwu, Ogunlaja, Otuechere, & Gideon, 2017; Chauhan & Agarwal, 2010; Koriem, 2013). For instance, fenugreek seed-supplemented diet was reported to induce male infertility by targeting the testes. It reduced sperm concentration and altered the testicular seminiferous tubules and the interstitial tissues of rabbits (Kassem, Al-Aghbari, AL-Habori,

& Al-Mamary, 2006). Additionally, *Momordica charantia* seed extract also distorted the structure of seminiferous tubules which was associated with considerable depletion in rat spermatogenesis (Eboetse et al., 2011). The extract of *Phyllanthus niruri*, a medicinal plant used as an antimalarial agent, was shown to possess antifertility properties by reducing testosterone concentration and distorting the seminiferous tubules of rats (Asare et al., 2013). Furthermore, the extract of *Xylopiya aethiopica* seeds used in folklore medicine was also reported to alter sperm parameters and testosterone levels in rats (Abarikwu, Ogunlaja, et al., 2017).

Telfairia occidentalis (TO), routinely referred to as fluted pumpkin, fluted gourd and "Ugu," is a popular plant and food item amongst the populace in Nigeria due to its consumable seeds and edible leaf. It is an integral part of folkloric medicine in Nigeria (Adisa, Okhai, Bankole, Iyamu, & Aigbe, 2014; Akoroda, 1990). In the eastern part of Nigeria, fleshy part obtained from seeds of boiled and shelled TO is often consumed as snacks or ground for inclusion in soup (Ajayi, Oderinde, Taiwo, & Agbedana, 2004; Akoroda, 1990). In fact, boiled TO seeds can be left to ferment, and the ensuing substance referred to as "Ogiri" serves as seasoning for soups and other foods (Akoroda, 1990; Saalu et al., 2010). The seeds are rich in phytosterols, phytochemicals, sterols, polyunsaturated fatty acids, tocopherol, carotenoids and proteins (Aghaei et al., 2013). Flavonoids, alkaloids, saponins, phenolics, anthraquinones and tannins were also found in methanolic seed extract of TO (Osukoya, Adegbenro, & Onasanya, 2016). Indeed, oil-rich seeds of TO are a veritable source of proteinous cake and unsaturated oils, while the leaves are sought after as potherbs, thus making TO an indispensable dietary component of traditional medicine (Akoroda, 1990). More so, it has been opined that bioactive constituents of TO can trigger varied physiological, biochemical and morphological effects (Saalu et al., 2010), and this may account for the divergent influence of TO in experimental models. For instance, the consumption of intact concentrate of TO seeds was accompanied with diminution of symptoms of benign prostate hyperplasia in experimental models (Abdel-Rahman, 2006). Pumpkin seed extract was also reported to have improved sperm characteristics in cyclophosphamide-treated rats (Aghaei et al., 2013), whereas the aqueous extract of TO leaves was reported to induce testicular damage in rats (Adisa et al., 2014; Saalu et al., 2010). Despite the extensive consumption of boiled TO seeds in eastern Nigeria, information on the effect of cooked fleshy TO kernel on sperm functionality, testicular structure and oxidative status of the testes is scarce. Thus, in the light of the above background, this work aims to determine the sperm quality, testosterone level and testicular oxidant-antioxidant parameters of rats consuming boiled TO-supplemented diet.

2 | MATERIALS AND METHODS

2.1 | Composition of diet

Telfairia occidentalis seeds were purchased from a local market in Rivers state, Nigeria. These seeds were certified by Dr. Ekeke Chimezia of the Department of Plant Science and

Biotechnology, University of Port Harcourt, with a specimen number of UPH/P/096. Seeds from mature fruits of TO were removed, cleaned and shelled. Carefully selected seeds were boiled for 30 min-1 hr at 100°C. The boiled seeds were sun-dried and ground. A diet containing 10% (100 g TO/kg), 15% (150 g TO/kg) and 30% (300 g TO/kg) of the boiled and ground seeds and 90%, 85% and 70% standard feed, respectively, was fashioned, mixed meticulously and made into pellets. The formulated diet was dried and used as test chow. The dried and pelletised standard feed served as control.

2.2 | Animal model, treatments and collection of samples

Thirty inbred male albino rats of about 120–150 g weight were used for this study. These animals were divided into four groups of five rats each and sheltered in an aerated condition for a 12-h dark/light interval, provided with normal rat feed and water ad libitum. The ethical regulations on animal care and handling of the National Academy of Science were followed during experiments. After 1 week of acclimatisation, animals in group I were used as the control group and took normal pelletised rat chow. Rats in groups II, III and IV were treated with feed supplemented with 10%, 15% and 30% boiled *T. occidentalis* seeds, respectively, for 60 days.

After the treatment for 60 days, weights of the animals were recorded, and blood samples were obtained via ocular puncture prior to sacrifice of the rats by cervical dislocation. Testes from each rat were excised and weighed, and the left testis was blended in 0.1 M Tris-HCl buffer (pH7.4, ice-cold) to yield a 10% homogenate. Supernatant obtained from centrifuging the 10% testicular homogenate at 10,000 × g, 4°C for 30 min was used to evaluate oxidative stress biomarkers. Spermatozoa isolated from rat cauda epididymides were used for live/dead ratio (sperm viability), sperm count, morphology and motility.

2.3 | Epididymal sperm number, motility, viability and morphological assays

Cauda epididymis crushed in normal saline and sieved with a nylon mesh was used to obtain the epididymal sperm numbers. Neubauer hemacytometer was used to count the spermatozoa according to the method illustrated by Pant and Srivastava (2003). Motile spermatozoa sequestered from the epididymis were estimated visually at 400 × magnification. Three distinct areas in each sample were used to estimate sperm motility and the average was used as the endmost motility score and the data were expressed as percentages (Abarikwu, Adesiyani, Oyeloja, Oyeyemi, & Farombi, 2010). Part of sperm concentrate dropped on a glass slide was smeared out with another slide, fixed in 95% ethanol and stained with 1% eosin and 5% nigrosine, and the resulting film was used for viability and morphological estimation. Deformity in various areas of spermatozoa was estimated according to the procedure illustrated by Wyrobek and Bruce (1980).

2.4 | Biochemical assay and procedures

Malondialdehyde (MDA), an index of lipid peroxidation (LPO), was measured as stated by Ohkawa, Ohishi, and Yagi (1979). Briefly, the homogenates were precipitated with trichloroacetic acid (30% w/v) and centrifuged at 3000 g for 10 min. Then, thiobarbituric acid (0.75% w/v) was incorporated with the obtained supernatant, and the resulting mixture was then incubated for 1 hr in a boiling water bath. After incubation, the solution was cooled and the absorption of the supernatant was recorded at 532 nm and expressed as micromolar of MDA per gram of tissue. Reduced glutathione (GSH) was determined at 412 nm using the method described by Sedlak and Lindsay (1968) which is based on the reaction with Ellman's reagent (0.4% DNTB in 0.1M phosphate buffer, pH 7.4.). The GSH level was expressed as unit per milligram of protein. Superoxide dismutase (SOD) activity was measured as detailed by Misra and Fridovich (1972). Briefly, the reaction was started by the addition of 0.3 ml of freshly prepared adrenaline (0.01%) to the mixture containing 2.5 ml carbonate buffer (0.05M, pH 10.2) and 0.2 mL sample. The mixture was inverted quickly, and readings were taken immediately at 480 nm against a blank containing all the components except the enzyme source at 3 min of 30-s interval in a spectrophotometer. The activity was expressed as unit per milligram of protein. Myeloperoxidase activity, an index of inflammation, was determined according to the method described by Suzuki, Ota, Sasagawa, Sakatani, and Fujikura (1983). Briefly, samples (10 µl) were mixed with 80 µl of 0.75 mM H₂O₂ and 110 µl of 2.9 mM tetramethylbenzidine in phosphate buffer (pH 5.4), and mixtures were incubated at 37°C for 5 min. The reaction was stopped by adding 50 µl of 2M H₂SO₄ and absorbance read at 450 nm.

2.5 | Estimation of testosterone level

Serum-free testosterone was determined using an enzyme immunoassay kit (AccuBind Elisa Microwells, California, USA). Briefly, controls, specimens and standard were disbursed to suitable wells, accompanied by testosterone horseradish peroxidase reagent and antitestosterone reagent, before mixing thoroughly and incubating at 37°C for 90 min. Wells were then rinsed with deionised water, and substrate solution was dispensed into each well, gently mixed

and incubated for 20 min. The reaction was stopped with the "stop reagent" and the absorbance recorded at 450 nm. The minimum detection limit in the testosterone assay was 0.1 ng/ml. The cross-reactivity assay with other steroids tested showed 0.01%.

2.6 | Fourier transform infrared spectroscopy analysis (FTIR)

The extract was mixed with potassium bromide to minimise the scattering effect of large crystals, and infrared spectra of functional groups in the mixtures were then measured on a Fourier transform infrared (FTIR) spectrophotometer (model 8400S, Shimadzu Corporation, Kyoto, Japan).

2.7 | Histological assessment

A testis from each animal was fixed in Bouin's solution for 6 h and was subsequently moved to 10% buffered formalin. The formalin-fixed tissues were embedded in paraffin wax, and 5-µm-thick sections were cut with a rotary microtome. The sections were stained with haematoxylin and eosin for microscopy.

2.8 | Statistical analysis

Statistical evaluation of measured parameters was conducted with SPSS Statistics 17.0 (SPSS Inc., Chicago, IL) and data denoted as mean ± SD. All the data were analysed by ANOVA followed by Duncan's test as a post hoc test. *p* values less than .05 were considered to indicate statistical significance.

3 | RESULTS

3.1 | Body weight and organ weights

According to the results depicted in Table 1, increased body weight of rats in all groups were statistically nonsignificant (*p* > .05) across all groups. The absolute and relative testes weights of animals in 10% TO diet and 15% TO diet groups were also statistically unchanged (*p* > .05) compared to the control values, while the absolute and relative weight of the testes in 30% diet group was increased compared to control values (Table 1).

TABLE 1 Effect of *Telfaria occidentalis* seed-supplemented diets on body weight, absolute and relative organ weights of the testes of rats

Groups	Testis weight		Body weight (g)	
	Absolute (g)	Relative (g/100 gm)	Initial	Final
Control	1.79 ± 0.45	1.06 ± 0.21	122.0 ± 4.24	166.8 ± 18.30
10% TO diet	1.19 ± 0.54	0.73 ± 0.33	150.0 ± 5.88*	163.5 ± 11.93
15% TO diet	1.67 ± 0.62	1.08 ± 0.46	140.4 ± 17.16*	156.5 ± 18.54
30% TO diet	2.52 ± 0.44*	1.39 ± 0.29*	133.2 ± 15.46	183.6 ± 19.91

Data are presented as the mean ± SD (*n* = 5).

*Significantly different from control group (*p* < .05).

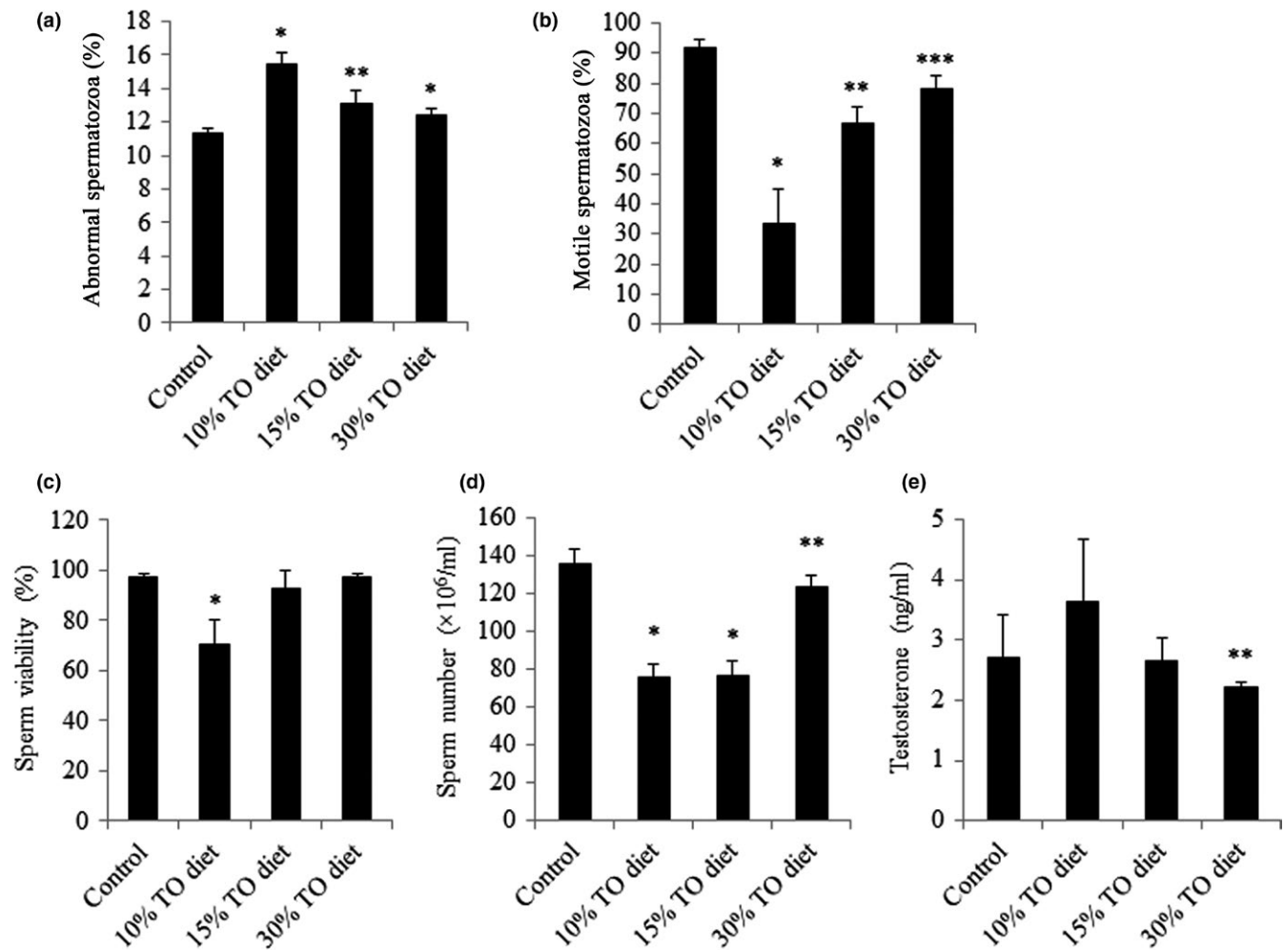


FIGURE 1 Effects of TO-supplemented diet on some sperm quality variables and serum testosterone level (a) abnormal spermatozoa (b) motile spermatozoa, (c) sperm viability (d) sperm number in the epididymides, (e) testosterone level. Data are presented as the mean $\pm$ SD (n = 5). *Significantly different from control group, **Significantly different from 10% TO group, ***Significantly different from 15% TO group (p < .05)

3.2 | Sperm quality of rats exposed to TO diets

Consumption of TO-supplemented diet at 10%, 15% and 30% significantly diminished (p < .05) sperm motility and number of spermatozoa in the epididymis, whereas the increased abnormal sperm morphology and dead spermatozoa observed in all TO diet groups were significant compared to control rats. However, motile spermatozoa and the number of spermatozoa in the epididymis were more reduced in the 10% TO diet groups than were observed in 15% and 30% TO diet groups (Figure 1). Furthermore, sperm viability was reduced significantly (p < .05) in rats fed 10% TO-supplemented diet as compared to rats in control group but remained unchanged in groups exposed to 15% and 30% TO diets compared to control animals (Figure 1).

3.3 | Concentration of serum testosterone and activity of serum MPO in rats administered TO diets

The elevation in serum testosterone in 10% and 15% TO diet groups was not statistically significant compared to control. However, 30%

TO augmented diet provoked a significant diminution (p < .05) in serum testosterone level when compared to control rats (Figure 1). The decrease in MPO activity in the serum of rats consuming 10% and 15% TO diets was not statistically significant (p > .05) in relation to control animals. Also, the elevated MPO activity observed in animals consuming 30% TO diet was not statistically significant (p > .05) comparable to control (Figure 2).

3.4 | Testicular SOD activity, and concentrations of GSH and MDA of rats consuming TO diets

In Figure 2, the increased testicular SOD activity and concentrations of GSH in rats consuming 10%, 15% and 30% TO-supplemented diets were not statistically significant (p > .05) as compared to values of control animals. Furthermore, consumption of 10% TO augmented diet significantly elevated (p < .05) the testicular MDA concentration in relation to control values, whereas there was increase in the concentration of MDA in 15% and 30% TO diet group which was not statistically significant compared to values of control (Figure 2).

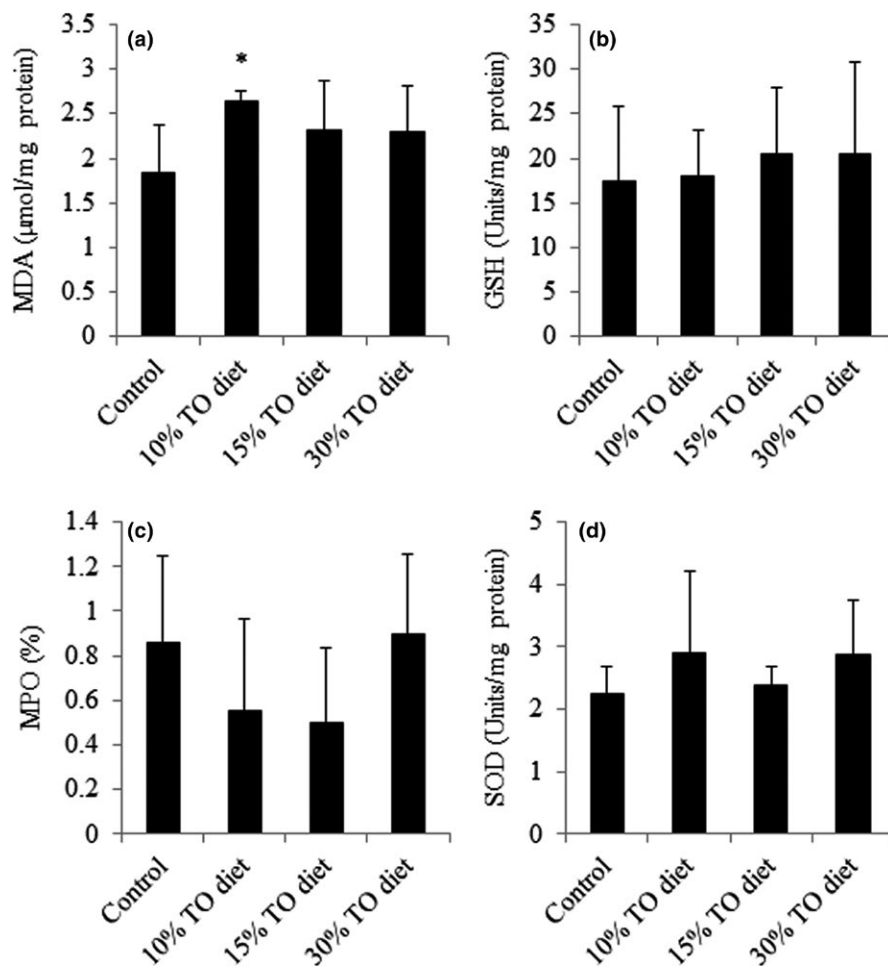


FIGURE 2 Effects of TO-supplemented diet on some biochemical parameters in the testes (a) malondialdehyde (MDA), (b) reduced glutathione (GSH), (c) myeloperoxidase (MPO), (d) superoxide dismutase (SOD). Data are presented as the mean $\pm$ SD ($n = 5$). *Significantly different from control group, ($p < .05$)

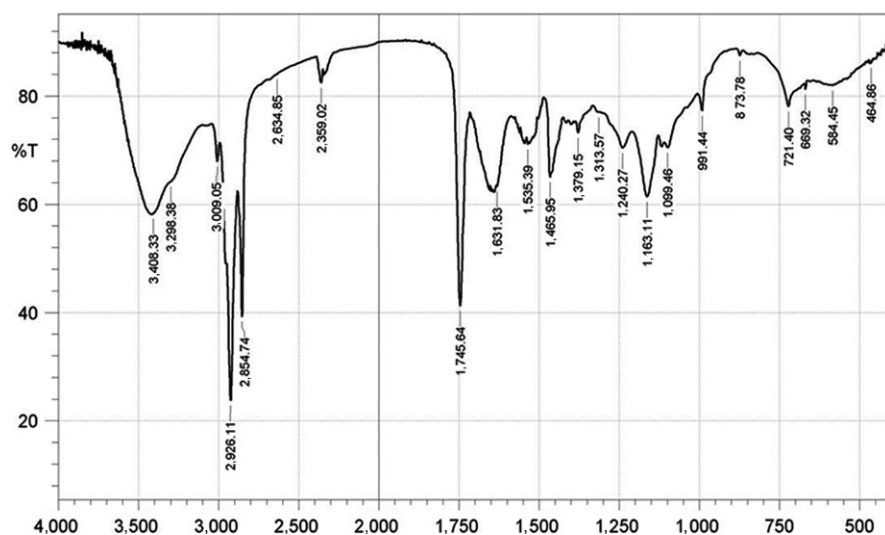


FIGURE 3 FTIR spectra of boiled TO showing OH, C-H, C=O and C=C absorbance peaks at $3,408\text{ cm}^{-1}$ and $3,298\text{ cm}^{-1}$, $2,926\text{ cm}^{-1}$ and $2,854\text{ cm}^{-1}$, $2,359\text{ cm}^{-1}$ and $1,745\text{ cm}^{-1}$ respectively

3.5 | FTIR characterisation of boiled TO seeds

The FTIR spectra indicated the significant presence of OH stretching at $3,408\text{ cm}^{-1}$ and $3,298\text{ cm}^{-1}$ and the presence of C-H of aliphatic hydrocarbon at $2,926\text{ cm}^{-1}$ and $2,854\text{ cm}^{-1}$ (Figure 3). The spectra

also depict C=O stretch at $2,359\text{ cm}^{-1}$, ester (C=O) vibration and alkene (C=C) at $1,745\text{ cm}^{-1}$ and $1,631\text{ cm}^{-1}$, respectively, while the aliphatic and aromatic C=C stretch were visible at $1,535\text{ cm}^{-1}$ and $1,465\text{ cm}^{-1}$ (Figure 3). These peaks suggest that the TO seeds contain a mixture of aromatic and aliphatic compounds.

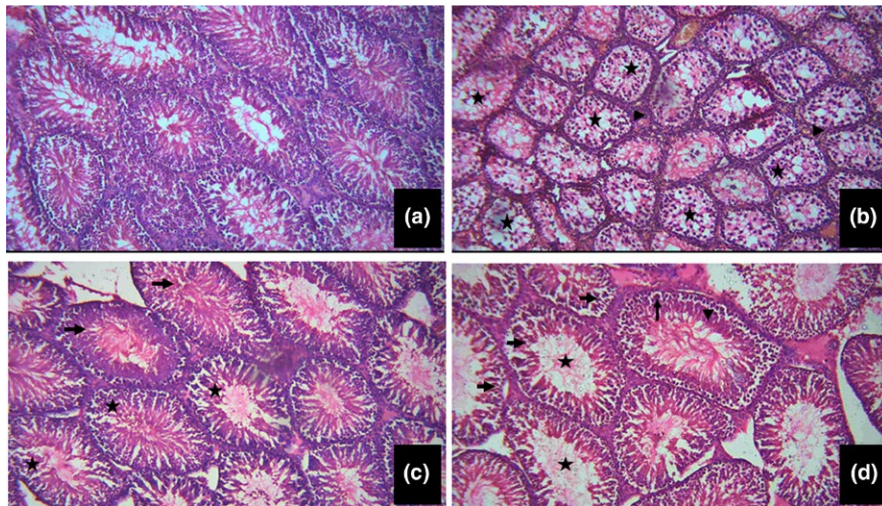


FIGURE 4 Representative photomicrograph of control and treated testes. (a) Control: No visible lesion seen (b) 10% TO diet: severe loss of seminiferous tubular germ cells (starred) and increased proliferation of Leydig cells in the interstitium (arrow head) (c) 15% TO diet: degeneration of seminiferous epithelium (starred), loss of most of the matured germ cells (short arrows) (d) 30% TO diet: moderate loss of germ cells (arrow head), extended lumen (starred), degenerated epithelium (short arrows), giant cell (long arrow). Haematoxylin and Eosin, Magnification 100 ×

3.6 | Histology of the testes

Figure 4 depicts the representative photomicrograph of sections of treated and control testes. There was no visible lesion in the testes of control rats. However, there was distortion of the seminiferous tubules accompanied by massive loss of germ cells and increased proliferation of Leydig cells in the testes of rats exposed to 10% TO diet. There were moderate loss of germ cells and degeneration of the seminiferous epithelium in the 15% and 30% TO diet groups.

4 | DISCUSSION

It is well known that diet modifies the physiological and pathological processes of spermatogenesis due to the influence of constituent phytochemicals and nutritional factors (Adisa et al., 2014; Ricci et al., 2018). Interestingly, data from the current study agree with the observations of Adisa et al. (2014) that TO negatively affects parameters of semen functionality. Increased body weights of rats in all groups, during the 60 days exposure to TO diets, indicates the absence of general toxicity (Abarikwu, Olufemi, et al., 2017). The nonsignificant decrease in relative and absolute testes weights of rats consuming 10% and 15% TO-supplemented diets could be due to mild aberrant morphology (Abarikwu, Njoku, Lawrence, Charles, & Ikewuchi, 2017) and also indicates that boiled seeds of TO are toxic to the interstitial tissue and seminiferous tubules (Adisa et al., 2014; Kassem et al., 2006). Increased abnormal sperm morphology and reduced epididymal sperm number were also observed in rats administered TO diets. The percentage of motile spermatozoa were reduced in all groups consuming TO diets, while feeding boiled TO seeds to rats at 10% significantly lowered sperm viability comparable to control values. Overall, poor sperm quality observed in this study was more prominent in rats fed 10% TO seed diet than other TO diet groups. Sperm production and quality are associated with architectural integrity of the testis (Saalu et al., 2010). The observed

effect could be attributed to the presence alkaloids, saponins and tannins (Adisa et al., 2014). Furthermore, metabolites resulting from bioactivation of these compounds are capable of binding to cell molecules and cross-linking DNA thereby causing cytotoxicity (Badifu & Ogunsua, 1991). Indeed, saponins isolated from medicinal plants exhibited antispermatic potential in experimental models. These natural surfactants interact with plasma membrane of spermatozoa thereby promoting vacuolation, vesiculation or dissolution of head region of acrosome (Dhar, Bajpai, Setty, & Kamboj, 1989). Additionally, alkaloid-rich plants have also been shown to exhibit spermicidal and antifertility potential at low doses (Rao, 1987). Thus, we suggest that TO diet elicits spermicidal effects at a lower dose due to a direct action on the seminiferous tubules and spermatozoa.

Several studies have reported low testosterone level associated with decline and/or derangements in male fertility parameters that were triggered by exogenous compounds. It is believed that viable concentrations of testosterone are required for commencement and continuation of spermatogenic process and a reduction in the level of androgen biosynthesis provokes aberrant sperm function (Abarikwu, Olufemi, et al., 2017). Sperm quality parameters are always reported to be poor in experimental models where testosterone level is low (Abarikwu, Ogunlaja, et al., 2017; Adisa et al., 2014; Eboetse et al., 2011; Kassem et al., 2006). Interestingly, in our study, rats fed with TO diets at 10% and 15% showed a nonsignificant increase in testosterone level, while rats exposed to 30% TO diet had lower concentration of testosterone. The increased level of testosterone in 10% and 15% TO diet groups is due to the abundance of Leydig cells in the interstitial areas of the testes of these animals. This agrees with earlier studies which demonstrated that exposure of experimental animals to testes damaging agents are associated with increased testosterone levels and poor semen quality (Haouem, Najjar, Hani, & Sakly, 2008; Zeng, Jin, Zhou, & Nordberg, 2003). Meistrich and Shetty (2003) had suggested that elevated levels of testosterone inhibit differentiation and promote apoptosis of germ cells via a direct

insult on the spermatogonia or androgen-responsive somatic cell. The scope of this study does not further allow us adduce the underlying mechanism for TO-associated elevation in testosterone level. However, it is suggested that TO at low doses could promote loss of germ cells in the seminiferous tubules due to the high testosterone levels, whereas the decrease in testosterone level in the 30% TO diet group is attributed to the few numbers of Leydig cells in the testis of these group of animals. Histologically, the aberrant impact of TO seed diet on the functional and structural integrity of the testes was shown by the degeneration of the seminiferous epithelium and loss of most of the germ cells (spermatogonia, spermatocytes and spermatids). This was more pronounced in the 10% TO diet group compared to other groups and was accompanied with large populations of Leydig cells in the interstitium of the 10% TO diet group. This allows us to also infer that feeding TO diet to rats resulted in a direct toxic insult to germ cells in the seminiferous tubules.

The elevation and/or reduction in antioxidant defence systems accompanied by a concomitant increase in MDA concentration highlights the induction of oxidative stress (Abarikwu, Njoku, et al., 2017; Imamoğlu, Bülbül, Kaklikkaya, & Sarihan, 2012). More so, the increase in the activities or levels of antioxidant defence in experimental models is seen as an adaptive response to oxidative stress. Animals that received TO diet showed increased activity of SOD as well as increased concentrations of GSH and MDA. However, the effects of TO diet on the investigated oxidative stress parameters appear to be mild. The elevated concentration of MDA was more prominent in the 10% TO diet group compared to control value, but the increase in MDA was not significant in the 15% and 30% TO diet groups and was associated with non-significant elevation in SOD activity and GSH level. TO-induced perturbations in testicular oxidative status noted in these investigations agree with the observations of Saalu et al. (2010) and could be attributed to peroxidative stress in testicular tissues induced by phytochemicals present in the plant (Hur & Gray, 2011; Iwalewa et al., 2005; Saalu et al., 2010). Spermatogenic process is vulnerable to testicular inflammation, which results in erosion of seminiferous epithelium and increased loss of spermatogenic cells (Imamoğlu et al., 2012). Thus, we also evaluated serum myeloperoxidase (MPO) activity, an enzyme discharged during inflammation and testicular damage (De Freitas et al., 2012). The nonsignificant increase of MPO activity in 30% TO diet group and reduction in MPO activity in animals fed 10% and 15% TO diets suggest that boiled TO seeds induce inflammatory changes in the testes of treated animals. The FTIR analysis and the spectra of the samples revealed the presence of alkenes, esters, phenyl and alcohol functional groups (Osho, Otuechere, Adeosun, Oluwagbemi, & Atolani, 2015; Owolabi et al., 2018). This agrees with earlier reports on the bioactive constituents of pumpkin seeds (Aghaei et al., 2013; Eseyin, Sattar, & Rathore, 2014). More so, OH, CH₃ and C=O functional groups found in alkaloids, and saponins of some plant products have been demonstrated to have antifertility properties (D'Cruz, Vaithinathan, Jubendradass, & Mathur,

2010; Singh, Dubey, Tripathi, Singh, & Saraf, 2014). It is therefore posited that antifertility effects exhibited by boiled TO seeds could be due to the influence of toxic phytoconstituents in TO or reactive metabolites arising from the transformation of these constituents.

5 | CONCLUSION

These data suggest that boiled TO seed-supplemented diets have spermatotoxic effects and alters testosterone profiles in rats. The potentials of TO seeds to provoke these changes in sperm quality are related to the presence of some compounds such as saponins and alkaloids in TO that have antifertility properties.

ACKNOWLEDGEMENTS

The technical expertise of Messrs Omoko Ejoro of the Department of Veterinary Surgery and Reproduction, Faculty of Veterinary Medicine, University of Ibadan, Ibadan, Nigeria, on the collection, handling and analysis of spermatozoa is gratefully acknowledged.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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How to cite this article: Njoku R-CC, Abarikwu SO, Uwakwe AA, Mgbudom-Okah CJ. *Telfairia occidentalis*-supplemented diet induces changes in sperm parameters and testosterone level in rats. *Andrologia*. 2018;50:e13044. <https://doi.org/10.1111/and.13044>